



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018**

CANDIDATE
NAME

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H1 BIOLOGY

Paper 2 Structured Questions

8876/02

**Friday 24 August 2018
2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, Centre number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use			
Q1	/ 6	Q5	/ 8
Q2	/ 10	Q6	/ 7
Q3	/ 5	Q7 / 8	/ 15
Q4	/ 9		
Total		/ 60	

This document consists of **16** printed pages.

Answer **all** the questions in this section.

1 Sugar molecules enter cells through transport proteins.

(a) Explain why transport proteins are required for the movement of sugar molecules, such as glucose and fructose, into cells.

[2]

Some plant cells convert fructose and glucose into sucrose for transport from the leaves to the roots. Sucrose is moved into phloem sieve tubes as shown in Fig. 1.1.

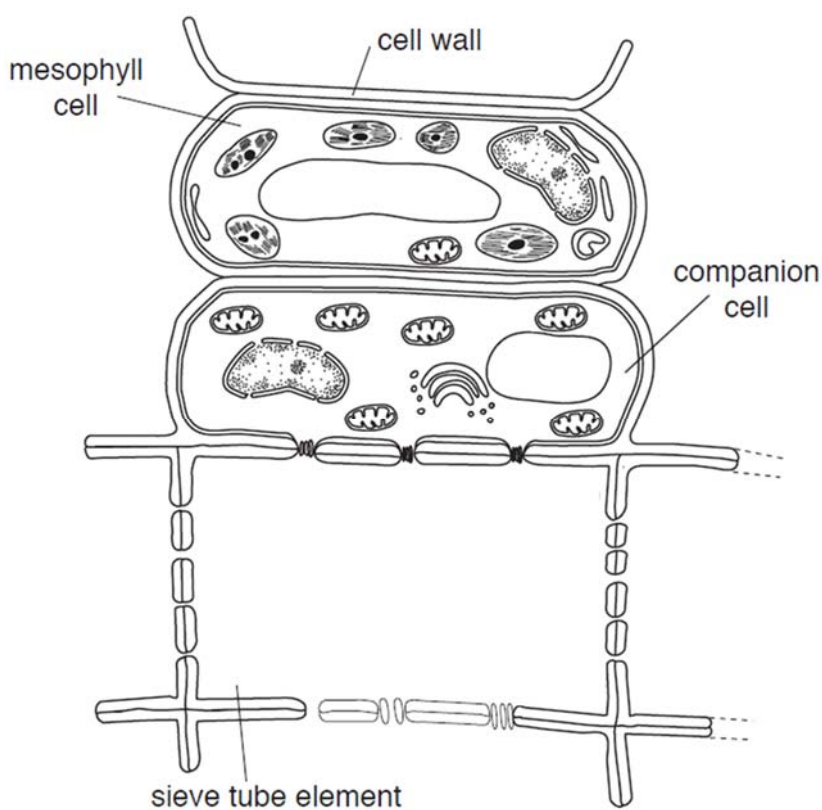


Fig. 1.1

Each cell has a specialized function.

(b) With reference to Fig. 1.1 and the information provided, state **one** difference between a mesophyll cell and companion cell.

[1]

Fig. 1.2 shows how sucrose is transported into the companion cell from mesophyll cell.

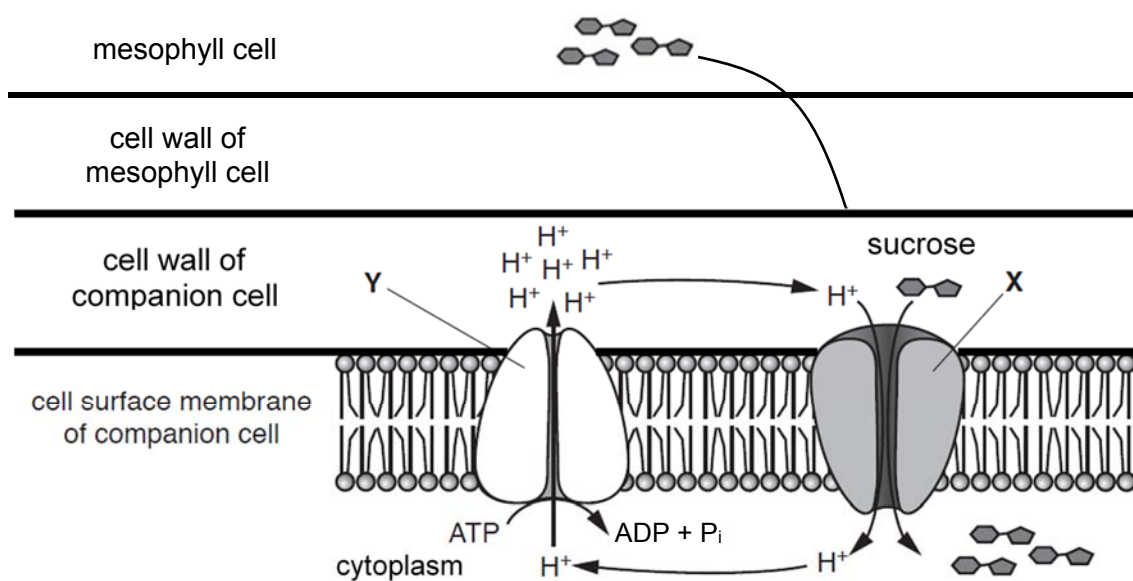


Fig. 1.2

(c) Using the information in Fig. 1.1 and Fig. 1.2, explain how sucrose moves into the companion cell.

[3]

[Total: 6]

- 2 The yeast, *Saccharomyces cerevisiae*, is a single-celled, eukaryotic organism that is often used in the laboratory.

When yeast is mixed with glucose solution, the yeast absorbs the glucose. Each molecule of glucose is then broken down into pyruvate molecules in exactly the same way as in any other eukaryotic organism.

- (a) Outline the breakdown of glucose to pyruvate in this stage.

[2]

Yeast cells sometimes carry out anaerobic respiration. Fig. 2.1 outlines the process of anaerobic respiration in yeast cells.

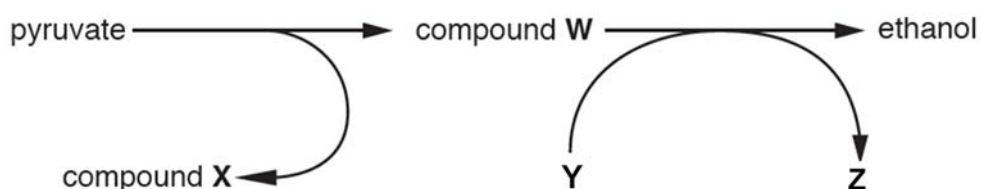


Fig. 2.1

- (b) (i) Identify molecule Z.

[1]

- (ii) State why molecule Y is converted to Z.

[1]

Yeasts are often used in bread-making. The bread dough is kneaded to introduce and trap air so that the yeasts in the dough can respire aerobically. Besides carbon dioxide that is released during respiration, the evaporation of water or ethanol released during respiration also causes the dough to rise.

Table 2.1 shows the differences in the height of dough that was placed at different locations, after the dough was kneaded.

Table 2.1

Time/ min	Height of dough/ cm		
	Fridge	Room temperature	Next to window (hot day)
0	2.5	2.5	2.5
20	2.5	2.9	3.3
40	2.7	3.7	4.0
60	2.9	3.9	4.7
80	3.0	4.0	5.2
100	3.0	4.0	5.8
120	3.0	4.0	6.0

- (c) (i) Account for the difference in the overall increase in the height of dough that was placed in the fridge with that placed next to the window.

[4]

- (ii) Suggest why the increase in the height of dough that was placed at room temperature was higher between 0 and 40 minutes than between 40 minutes and 60 minutes.

[2]

- 3 Fig. 3.1 shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

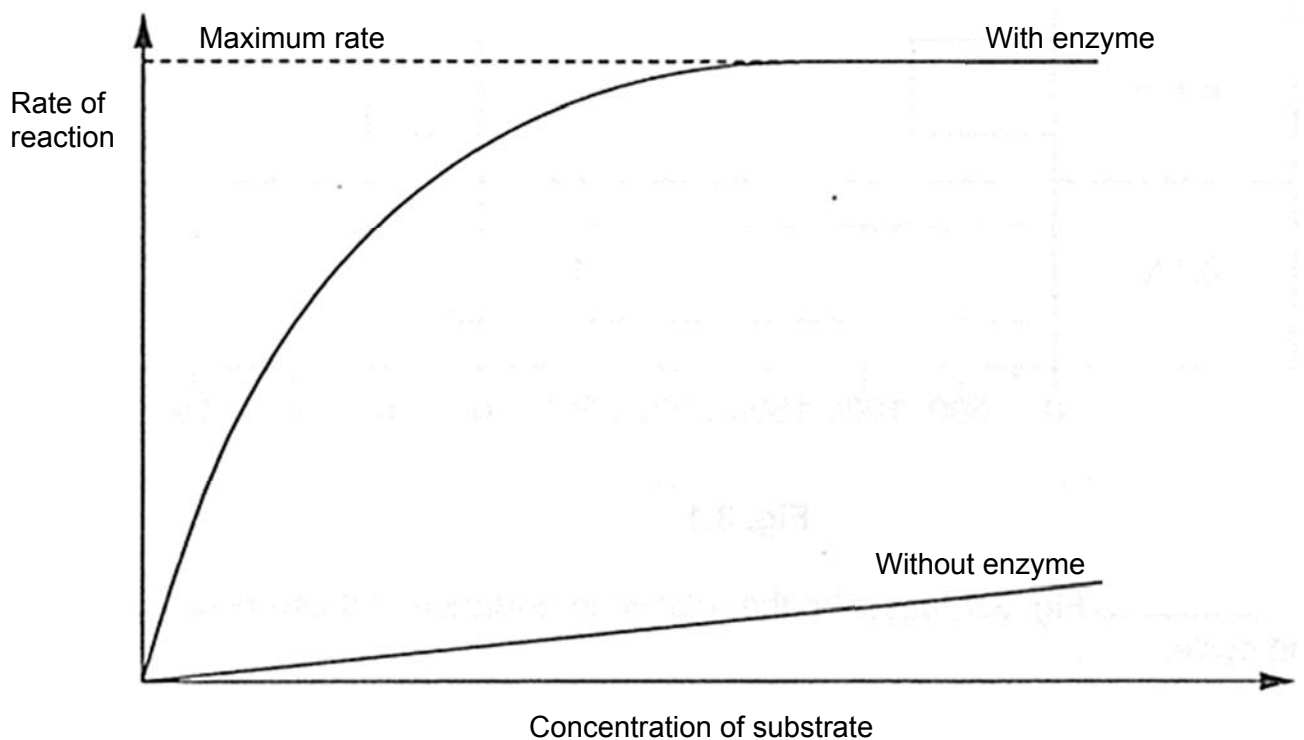


Fig. 3.1

- (a) On Fig. 3.1, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction upon the addition of

- (i) a competitive inhibitor;
- (ii) a non-competitive inhibitor.

[2]

- (b) Explain the effect of a competitive inhibitor on the rate of enzyme activity.

[3]

[Total: 5]

- 4 Table 4.1 shows some of the common fatty acids and their melting points.

Table 4.1

Symbol (number of carbon atoms : number of double bonds)	Common Name	Melting point (°C)
<i>Saturated fatty acids</i>		
12 : 0	Lauric acid	44.2
14 : 0	Myristic acid	52
16 : 0	Palmitic acid	63.1
18 : 0	Stearic acid	69.6
20 : 0	Arachidic acid	75.4
22 : 0	Behenic acid	81
<i>Unsaturated fatty acids</i>		
16 : 1	Palmitoleic acid	-0.5
18 : 1	Oleic acid	13.4
18 : 2	Linoleic acid	-9
18 : 3	α -linolenic acid	-17
20 : 4	Arachnidonic acid	-49.5

- (a) Arachidonic acid is a polyunsaturated fatty acid.

Explain the term *polyunsaturated fatty acid*.

[1]

- (b) With reference to Table 4.1,

- (i) describe the effect of increasing number of carbon atoms in saturated fatty acids on the melting point;

[3]

(ii) describe the effect of the presence of double bonds in fatty acids on the melting point;

[1]

(iii) explain the trend described in **b(ii)**.

[4]

[Total: 9]

- 5 Table 5.1 provides statements regarding the bonds found in four biological molecules.

Table 5.1

statement	protein	DNA	messenger RNA	cellulose
hydrogen bonds stabilise the molecule				
subunits are joined by peptide bonds				

- (a) Complete Table 5.1 by indicating with a tick (✓) or a cross (✗) whether the statements apply to proteins, DNA, messenger RNA and cellulose.

You should put a tick or a cross in each box of the table.

[2]

- (b) A piece of mRNA is 660 nucleotides long but the DNA coding strand from which it was transcribed is 870 nucleotides long.

- (i) Explain this difference in the number of nucleotides

_____ [1]

- (ii) What is the maximum number of amino acids in the protein translated from this piece of mRNA? Explain your answer.

Number of amino acids _____

Explanation

 _____ [2]

- (c) Identify **one** other process that lead the formation of mature mRNA and state its function.

 _____ [2]

- (d) Describe **one** difference between the structure of mRNA and tRNA.

 _____ [1]

[Total: 8]

The comparisons with three species of amphibians, *Xenopus laevis* (Xl), *X. tropicana* (Xt) and *Rana catesbeiana* (Rc) are shown in Table 6.1.

		percentage of matches of amino acid sequence					
		species of amphibian adults			species of amphibian larvae (tadpoles)		
	fish species	<i>Xl</i>	<i>Xt</i>	<i>Rc</i>	<i>Xl</i>	<i>Xt</i>	<i>Rc</i>
α chains	coelacanth	42.0	47.5	no data	45.4	42.6	48.2
	lungfish	40.4	42.1	no data	40.7	39.0	37.9
β chains	coelacanth	42.1	43.2	40.7	52.1	52.1	58.2
	lungfish	44.1	45.9	41.4	47.3	45.9	48.6

[4]

- (b) Coelacanth haemoglobin has a very high affinity for oxygen, suggesting that coelacanths, which have been captured at depths of between 200 m and 400 m, live in water that has a low concentration of oxygen.

Explain how an environmental factor, such as the low concentration of oxygen in deep water, can act as an evolutionary force in natural selection.

[3]

[Total: 7]

✂ End of Section A ✂

